

Leveraging Vision Foundation Models with a Lightweight PyraFuse Decoder for Skin-Fabric Semantic Segmentation

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Abstract—Pixel-level separation of skin from fabric is a core requirement for virtual try-on, privacy-preserving video analytics, and dermatology applications. Existing work treats skin detection and garment parsing as separate problems: skin detectors return a binary skin/non-skin map, while fashion parsers fold exposed skin into the background, so neither yields the three-way decision these applications require. We address three-class (background, fabric, skin) segmentation by pairing a frozen Vision Foundation Model (VFM) encoder with a lightweight decoder. We build a full pipeline from dataset construction through TensorRT deployment, including class-imbalance handling, and we propose PyraFuse, a lightweight pyramid decoder combining LayerNorm-aligned token branches, SE-gated depthwise blocks, a learned per-channel fusion gate, and a pyramid-pooling context module. A frozen DINOv3 encoder with PyraFuse reaches 92.1% mean IoU (98.5 background, 95.0 fabric, 82.9 skin) on a 4,594-image test split. We further report a skin-tone fairness audit and a precision-throughput sweep that assess the design under deployment conditions. Code and dataset-preparation scripts are available at <https://github.com/jamal-saeedi/PyraFuse>.

Keywords—*semantic segmentation, skin detection, vision foundation models, frozen backbone, lightweight decoder, TensorRT, real-time inference.*

I. INTRODUCTION

Pixel-accurate separation of skin from worn fabric against an arbitrary background underpins a growing set of practical systems, among them virtual try-on and garment retexturing, body measurement, sun-exposure and cosmetics analysis, privacy filters that selectively blur exposed skin in care-monitoring video, and content moderation. For all this breadth of application, the joint skin-versus-fabric problem is rarely posed as a single task. Skin detection is usually framed as a binary skin/non-skin decision, frequently with color-space heuristics that break down under the wide variation in lighting, skin tone, and skin-colored clothing that fashion imagery presents. Fashion parsing, in turn, segments garments into fine categories, yet labels exposed skin as background. A system that must decide whether a pixel is skin, fabric, or neither is therefore left without either an off-the-shelf dataset or an established encoder-decoder recipe, and this work addresses that gap with a deployment-oriented design.

The proposed method has three parts: a frozen VFM encoder, a small decoder trained on its features, and a deployment path covering dataset merging, imbalance

handling, augmentation, and export. Freezing the backbone keeps the study on a single GPU. One encoder serves every decoder revision and is compiled to TensorRT once. The remainder of the paper describes each stage of the pipeline, from data preparation to deployment, with a controlled experiment behind every design decision.

This work makes two primary contributions. The first is a reproducible end-to-end pipeline for joint skin-fabric segmentation. It comprises a documented procedure for merging visuAAL skin masks with Fashionpedia garment instances into a coherent three-class corpus of 44,783 training images, complete with conflict-resolution rules and pixel-level class-balance estimation; a training protocol that copes with severe imbalance (roughly 16:1 background-to-skin) through tone-preserving augmentation; and a complete ONNX/TensorRT showcase whose FP16/INT8 precision sweep delivers 300 images per second at negligible accuracy loss. The second is PyraFuse, a pyramid decoder with approximately 1.25 M parameters; it combines LayerNorm-aligned token branches, SE-gated depthwise blocks, a learned per-channel fusion gate, and a lightweight pyramid-pooling context module [1]. In a 42-run benchmark spanning seven encoders and six decoders, PyraFuse achieves the best result on six of the seven encoders, outperforming DPT, UPerNet, TPA-SAD, Mask2Former, and SegFormer heads at comparable or lower parameter cost.

II. RELATED WORK

A. Skin Detection and Segmentation

Classical skin detection thresholds color in YCbCr or HSV. Such methods are fast but fragile: they degrade under changing illumination and skin tone, and they cannot tell skin apart from skin-colored fabric. Convolutional models improved robustness, and the visuAAL Skin Segmentation dataset [2] supplied large-scale skin masks aimed at privacy-preserving applications. Surveys of the area [3] single out class imbalance and skin-tone bias as the principal open problems. In this work, we address both issues, the former through inverse-frequency loss weighting and the latter through tone-preserving augmentation, and we audit the residual bias directly in Section VI.

B. Fashion and Human Parsing

Fashion benchmarks such as ModaNet and Fashionpedia [4] segment garments into fine-grained categories with

instance polygons, whereas human-parsing datasets such as LIP and CIHP jointly label body parts and clothing, though at a granularity that does not align with a simple skin-fabric split. We use Fashionpedia as the fabric source because its apparel categories map naturally onto a single fabric superclass, and its instance masks are of high quality. Since no public benchmark targets the three-class background-fabric-skin formulation directly, we construct the merged dataset described in Section III.

C. Vision Foundation Models and Dense Features

DINO and DINOv2 [5] demonstrated that self-supervised ViTs yield patch features with emergent segmentation ability, usable while frozen behind a lightweight head. DINOv3 [6] scaled this idea using Gram anchoring to preserve dense feature quality, and its register tokens suppress the high-norm artifacts that contaminate DINOv2 attention maps [5]. AM-RADIO [7] distills several teachers, namely DINOv2, CLIP, and SAM, into one agglomerative backbone. SAM [8] produces promptable masks, but its features are tuned for class-agnostic boundaries rather than semantic labels. The language-supervised models, CLIP [9], SigLIP [10], and EVA-CLIP [11], carry rich semantics but a spatially coarser patch structure; for these, we recover localized patch-text similarity with a MaskCLIP-style value-only projection [12], which gives them a fair footing in the comparison of Section V.

D. Dense Decoders over ViT Features

A plain ViT emits a single-resolution patch grid, so any dense head must rebuild a multi-scale pyramid from it. DPT [13] addresses this by drawing tokens from several intermediate encoder depths and reassembling them through a convolutional pyramid, giving it a strong multi-scale representation but requiring a relatively heavy read-out and fusion path that scales with encoder depth. UPerNet [14] takes a complementary route, coupling a Feature Pyramid Network (FPN) with a pyramid-pooling module to capture both local detail and global context; the design is architecturally rich but inherits the cost of maintaining and fusing multiple FPN levels simultaneously. SegFormer [15] deliberately simplifies the decoder side, replacing all spatial operations with a flat MLP that merges features from several encoder stages; this keeps the head extremely compact but sacrifices the explicit spatial hierarchy that pyramid-based decoders maintain. Mask2Former [16] departs from the dense-prediction paradigm entirely, recasting segmentation as mask classification through a transformer decoder with cross-attention queries over learned object embeddings; the approach is the most expressive of these families and handles arbitrary category counts gracefully, but its attention over a full query set makes it the most computationally demanding.

Lightweight DPT-style variants instead target the frozen-encoder, small-head, real-time regime. Two representatives sit closest to our setting: SegDINO [17], which aligns multi-level frozen features to a shared resolution and predicts masks with an MLP head, and the Token-Pyramid Aligning plus Spatial Aggregation Decoder (TPA-SAD) [17], which organizes intermediate ViT tokens into a pseudo-pyramid before a lightweight spatial-aggregation decoder. This design

avoids heavy convolutional stacks and keeps parameter count minimal. PyraFuse operates in the same regime and shares the same frozen-encoder philosophy, but differs architecturally in three respects: it introduces LayerNorm-aligned projection branches for each pyramid level rather than relying on raw token reassembly, gates each depthwise convolutional block with Squeeze-and-Excitation (SE) channel attention to suppress uninformative activations, and replaces the fixed aggregation in TPA-SAD with a learned per-channel fusion gate [18] that dynamically weights level contributions at inference time. A lightweight pyramid-pooling context module further injects global receptive-field coverage without a separate heavy branch. Section V compares all of these families empirically over one identical frozen encoder.

III. DATASET AND TRAINING PROTOCOL

A. Sources and Merging Protocol

We merge two complementary sources. visuAAL Skin Segmentation [2] contributes roughly 46,000 images with binary skin masks that span diverse poses, illumination, and skin tones, while Fashionpedia [4] contributes roughly 48,000 fashion images with polygon instance masks over garments. The target label space is {0: background, 1: fabric, 2: skin} (Fig. 1). For Fashionpedia images, we rasterize every apparel instance and collapse it into the fabric class; because exposed skin is unlabeled at source, skin supervision is restricted to images where it is reliably available. For visuAAL images, the skin masks map directly to class 2, and the clothing regions supply fabric and background evidence. Where annotations overlap, as they do for sheer or skin-tight garments, we apply the priority rule skin over fabric over background inside any skin region, because a missed skin pixel is the costlier error for downstream privacy applications. After de-duplication and the removal of near-empty masks, the corpus holds 44,783 training, 1,157 validation, and 4,594 test images, split image-wise with seed 42 and no identity overlap.

B. Class-Balance Estimation

The pixel distribution is heavily skewed: background occupies 80.2% of pixels, fabric 14.7%, and skin only 5.1%, an imbalance of about 16:1 between background and skin. We estimate per-class frequencies once from the raw training

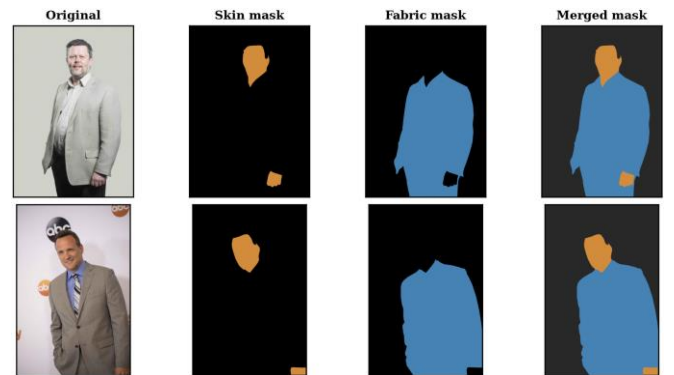


Fig. 1. Merged three-class annotations: image, skin mask, fabric mask, and merged mask, shown across skin tones and on a skin-colored-garment hard case.

masks rather than from augmented crops, which would bias the estimate toward the augmentation policy, then derive inverse-frequency weights and normalize them to sum to the number of classes C so that weighted and unweighted runs stay directly comparable. The resulting focal alpha is approximately [0.16, 0.52, 2.33] for background, fabric, and skin.

C. Preprocessing and Augmentation

Inputs are cropped to 448 by 448, a multiple of the patch size 16 that yields a 28 by 28 token grid, and normalized with ImageNet statistics to match backbone pretraining. Geometric augmentation consists of horizontal flips, random affine warps, and RandomResizedCrop, with nearest-neighbor interpolation on the masks so that labels do not bleed across boundaries. Photometric augmentation is intentionally aggressive in both HSV and RGB jitter: since the encoder is frozen, the decoder has to learn tone-invariant skin evidence from feature statistics alone, and heavy color augmentation proved to be the single most effective remedy for skin-tone bias (Fig. 2).

On top of this, we apply a skin-tone-specific transform, SkinToneShift, with probability 0.6 per batch. It perturbs the hue, saturation, and brightness of skin pixels only, with a downward brightness bias, 0.3, that over-generates the under-represented dark-tone examples corresponding to Fitzpatrick Types V and VI. Global hue shifts are bounded at plus or minus 15 degrees and are joined by a skin-targeted shift of plus or minus 8 degrees, which keeps skin within a plausible tone manifold while broadening the range of tones the model encounters. Together with the standard photometric transforms (brightness and contrast, RGB shift, gamma variation, and random shadows), this policy pushes the decoder toward color-invariant skin features instead of letting it memorize a single average skin color.

IV. ARCHITECTURE

A. Frozen-Backbone Protocol

Every experiment follows one protocol: the encoder stays frozen and only the decoder and classification head are trained. This holds the trainable parameter count to roughly 1.13 to 1.25 M, counting the decoder together with the per-scale adapter norms, so that the full 42-run benchmark fits on a single GPU. With no gradients flowing into the encoder, the backbone runs under *no_grad* and stores no activations for backpropagation, which lowers peak memory and allows a large batch at 448 by 448. The frozen encoder can also be exported to ONNX/TensorRT once and then reused across decoder revisions.

B. Encoder Comparison and Selection

We evaluate seven frozen backbones under a fixed decoder and training budget: DINOv3 ViT-B/16 (self-supervised with Gram anchoring), DINOv2 ViT-S/14 (self-supervised), AM-RADIO-B (multi-teacher distillation), SAM ViT-B (promptable segmentation), and the language-supervised CLIP ViT-B/16, SigLIP ViT-B/16, and EVA-CLIP ViT-B/16. All of them sit behind a common backbone adapter that taps hidden states at four depths and reshapes each into a [C,



Fig. 2. Augmentation pipeline preview: geometric transforms, photometric jitter, and the skin-tone-specific shift.

28, 28] spatial map. Because different VFMs emit feature maps whose per-channel scale and variance vary widely, a per-scale LayerNorm at the adapter boundary brings every map to a consistent distribution before the decoder sees it. This norm is the only trainable parameter inside the otherwise-frozen encoder path, and it keeps the head both encoder-agnostic and batch-size independent.

Two training-free diagnostics guide the analysis: PCA segmentation maps, which expose semantic cluster separability across all four tapped levels (Fig. 3), and center-pixel cosine-similarity maps, which probe fine-level discriminability (Fig. 4).

DINOv3 exhibits the clearest separation among the evaluated encoders: three separable PCA clusters that remain stable across levels (Fig. 3) and the tightest similarity falloff (Fig. 4). AM-RADIO ranks next, retaining coherent structure across all four levels. We attribute this to its multi-teacher distillation, where the DINO teacher contributes spatial coherence and the CLIP teacher adds semantic structure. DINOv2 separates the three regions well at every level and, thanks to its patch-14 configuration, offers a denser token grid that aids fine boundaries, which makes it the lightweight fallback. The language-supervised encoders (CLIP, SigLIP, EVA-CLIP) show spatially incoherent fine-level features, with neighboring patches on the same surface pointing in nearly orthogonal directions, and SAM cannot tell skin from fabric in the absence of a spatial prompt. Section V confirms these readings quantitatively.

C. PyraFuse Decoder

PyraFuse is a lightweight pyramid decoder built at the same roughly 1.25 M-parameter budget as TPA-SAD, but with four targeted refinements (Fig. 5). Each of the four backbone levels is projected to $dc = 128$ channels by a 1×1 convolution and then resampled by factors {8, 4, 2, 1} into a genuine spatial pyramid. The decoder incorporates four modifications relative to the baseline design.

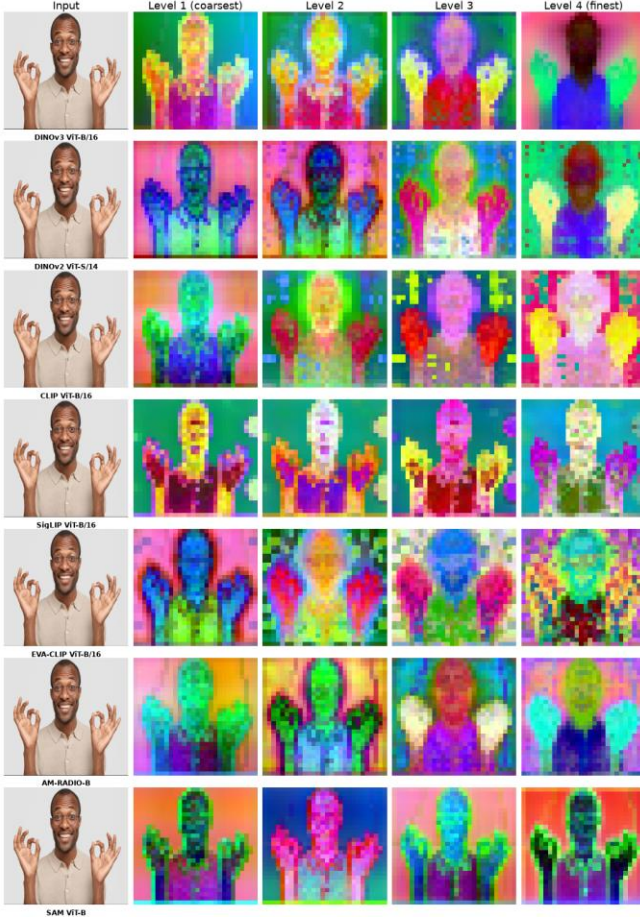


Fig. 3. Per-level PCA feature maps, showing the progression from high-frequency detail in early blocks to class semantics in late blocks.

1) Projection & Pyramid Construction

Let $F_i \in \mathbb{R}^{B \times C_{enc} \times H_i \times W_i}$ denote the feature map tapped from backbone level $i \in \{1, 2, 3, 4\}$, where $i = 1$ is the finest spatial level and $i = 4$ the coarsest. For a ViT-B/16 backbone processing a 448×448 input, the spatial resolution is $H_i = W_i = 28$ for all levels (the architecture is isotropic), and the embedding dimension is $C_{enc} = 768$. Each backbone feature map is independently projected to d_c channels via a 1×1 convolution and bilinearly resampled to the finest spatial resolution $H_1 \times W_1 = 28 \times 28$:

$$\hat{F}_i = \text{Interp}_{s_i} (W_i^{\text{proj}} * F_i), \quad s_i \in \{8, 4, 2, 1\} \quad (1)$$

where $W_i^{\text{proj}} \in \mathbb{R}^{d_c \times C_{enc} \times 1 \times 1}$ is a learned 1×1 convolutional kernel and Interp_{s_i} denotes bilinear upsampling by integer factor s . The four resulting maps $\{\hat{F}_i\}_{i=1}^4$ form a pyramid in which all levels share the same spatial resolution and channel count, making subsequent cross-scale operations parameter-free with respect to spatial alignment.

2) LN-Aligned Branches

Frozen ViT features carry per-channel scale heterogeneity that accumulates across depths. Before the intra-scale convolution, each projected branch is normalized and activated:

$$B_i = \text{Conv}_{3 \times 3}(\text{GELU}(\text{LN}(\hat{F}_i))) \quad (2)$$

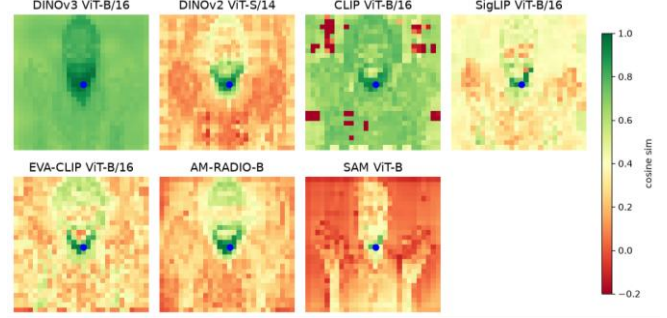


Fig. 4. Center-pixel cosine-similarity maps at the finest projected level for the seven frozen encoders. DINOv3 yields the sharpest, cleanest falloff; the CLIP-family encoders scatter low-similarity islands across uniform skin regions.

Here LN denotes LayerNorm applied independently at each spatial position over the channel dimension (token-wise), exactly matching the normalization used by the ViT internally. This statistical alignment means that all four projected levels present the same distributional profile to the subsequent convolutional kernel, regardless of which block depth they were extracted from. Without this step, the scale mismatch between early and late blocks degrades the quality of the cross-scale fusion gate.

3) SE-Gated Blocks

Intra- and inter-scale refinement uses residual depthwise-separable blocks fitted with a Squeeze-and-Excitation gate. Given input $x \in \mathbb{R}^{B \times d_c \times H \times W}$, the block computes:

$$z = \text{PWConv}_{1 \times 1}(\text{GELU}(\text{GN}(\text{DWConv}_{3 \times 3}(x)))) \quad (3)$$

$$\text{SE}(z) = z \odot \sigma(W_2 \text{ReLU}(W_1 \text{GAP}(z))) \quad (4)$$

$$\text{SEBlock}(x) = x + \text{SE}(z) \quad (5)$$

where $\text{PWConv}_{1 \times 1}$ is pointwise convolution and $\text{DWConv}_{3 \times 3}$ is a depthwise separable convolution [19], GN is GroupNorm with 8 groups, GAP is global average pooling, $\odot$ denotes channel-wise multiplication, $W_1 \in \mathbb{R}^{(d_c/r) \times d_c}$ and $W_2 \in \mathbb{R}^{d_c \times (d_c/r)}$ are the SE bottleneck matrices with reduction ratio $r = 4$, and σ is the sigmoid. The residual skip preserves gradient flow; the SE gate provides learned per-channel recalibration [20], replacing TPA-SAD's fixed scalar residual multiplier with a data-driven channel-attention weighting at negligible extra cost. The complete decoder uses eight instances of this block: four Intra blocks (one per pyramid level) and four Inter blocks (one per top-down fusion step).

4) Fusion Gate

Coarse-to-fine fusion at level $i \in \{3, 2, 1\}$ blends the upsampled coarser signal u^{i+1} (bilinearly upsampled by $\times 2$) with the intra-scale skip feature h_i (output of Intra_i):

$$g_i = \sigma(\alpha_i) \odot u_{i+1}^{\text{up}} + (1 - \sigma(\alpha_i)) \odot h_i \quad (6)$$

The fused result is then refined by the Inter block at that level, $u_i = \text{Inter}_i(g_i)$.

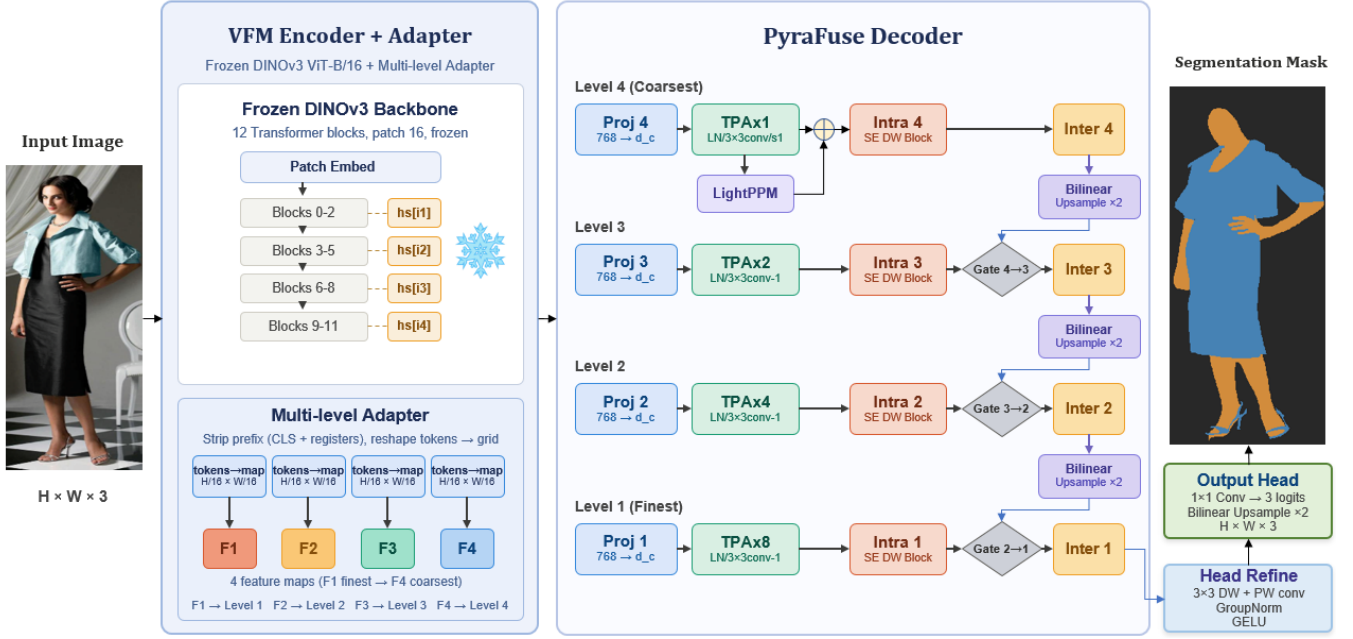


Fig. 5. PyraFuse architecture over a frozen DINOv3 ViT-B/16 encoder. Four tapped levels feed LayerNorm-aligned TPA branches, then SE-gated depthwise blocks, learned alpha-gated top-down fusion, and a LightPPM context module, before a refined 1 by 1 head produces the three-class mask

For the coarsest level there is no skip; the path begins with $u_4 = \text{Inter}_4(h_4)$. The fusion gate has several useful properties. First, the convex combination enforced by σ keeps the output in the convex hull of the two inputs, preventing gradient explosion from unconstrained addition. Second, the per-channel parametrization allows different channels to favor coarse context versus fine detail independently. Third, initialization at $\alpha = 0$ means training starts from a balanced 0.5 blend with gradients that are equal in both directions, avoiding the asymmetric initialization bias present in TPA-SAD's plain addition.

5) LightPPM

A lightweight Pyramid Pooling Module is attached to the coarsest branch B_4 before Intra_4 . For pooling scales $P = \{1, 2, 3, 6\}$, each branch pools, compresses, and upsamples back to the pyramid resolution:

$$p_k = \text{Interp}_{H1}(\text{Conv}_{1 \times 1}^{\frac{dc}{4}}(\text{AvgPool}_{p_k}(B_4))), p_k \in P \quad (7)$$

$$B_4^{\text{PPM}} = \text{Conv}_{1 \times 1}^{\text{dc}}([B_4; p_1; p_2; p_3; p_6]) \quad (8)$$

where $[;]$ denotes channel concatenation. Each pooling branch uses a $d_c/4 = 32$ -channel bottleneck. The aggregated map B_4^{PPM} replaces B_4 as input to Intra_4 , supplying receptive-field coverage of the full 448×448 input that the local 3×3 blocks in TPA-SAD cannot reach.

6) Head & Output

After top-down fusion produces u_1 at the finest level, a lightweight refinement block smooths the feature map before classification:

$$r = \text{PWConv}_{1 \times 1}(\text{GELU}(\text{GN}(\text{DWConv}_{3 \times 3}(u_1)))) \quad (9)$$

$$\hat{Y} = \text{Interp} \times_2(\text{Conv}_{1 \times 1}^{\text{ccls}}(r)) \in \mathbb{R}^{B \times 3 \times H \times W} \quad (10)$$

where $H \times W$ is the original input resolution. Logits are upsampled exactly once, at the end, rather than after each fusion step, which avoids repeated bilinear interpolation of class-specific responses and keeps the computation graph clean for TensorRT export. GroupNorm is used throughout the decoder, so the output distribution is independent of the inference batch size.

The backbone (DINOv3 ViT-B/16) accounts for approximately 35 G FLOPs at 448×448 , so the decoder represents less than 5% of the total inference compute. This is consistent with the throughput measurements in Section VII, where the encoder dominates end-to-end latency at every precision level.

GroupNorm with 8 groups is used throughout the decoder instead of BatchNorm for two reasons. First, BatchNorm statistics depend on the batch at inference time; GroupNorm does not, which ensures that the model's accuracy is identical whether it is run with batch size 1 (single-frame monitoring) or batch size 64 (offline processing). Second, TensorRT fuses GroupNorm into adjacent convolution kernels, whereas it cannot fuse BatchNorm when the mean and variance are dynamic, which gives a small but measurable latency benefit in the FP16 TensorRT engine.

D. Comparison with Other Decoders

Over frozen DINOv3-B features PyraFuse reaches 0.919 mIoU, ahead of DPT (0.909, about 2.8 M parameters), TPA-SAD (0.902, about 1.25 M), UPerNet (0.901, about 3.4 M), Mask2Former (0.892, heavy), and the all-MLP SegFormer-style head (0.888, minimal). In our experiments, Mask2Former was both slower and slightly less accurate than the proposed approach, which rules it out for the real-time target. Across the full 42-run grid PyraFuse is the best decoder

for six of the seven encoders, the lone exception being SigLIP, where DPT edges ahead by 0.01, and it beats TPA-SAD on every encoder. PyraFuse therefore remains the best decoder across nearly all encoders we tested (Fig. 6).

Figure 7 shows predictions on a set of randomly chosen test images. On this unbiased sample PyraFuse produces clean masks across different poses, garments, and lighting, with reliable skin/fabric separation and sharp boundaries. Since these images were not hand-picked, the results suggest the quantitative gains carry over to ordinary inputs.

V. DESIGN-SPACE STUDIES

We present the pipeline as an explicit sequence, encoder then decoder then loss then regularization, and support each choice with a controlled study over the 42-run grid of seven encoders by six decoders.

A. Encoder Selection

Table I reports the frozen-encoder comparison under an identical PyraFuse decoder, training budget, and augmentation. For the language-supervised encoders (CLIP, SigLIP, EVA-CLIP) we add a lightweight, training-free text-saliency gate (a MaskCLIP value-only projection of patch tokens, cosine-matched to fixed class prompts) residually, so as to re-weight their features toward class semantics. This evens out the comparison between vision-language and vision-only models; it is an evaluation aid, not a contribution.

DINOv3-B is the strongest encoder at 0.919 mIoU, with AM-RADIO (0.907) and DINOv2 (0.906) close enough to be statistically indistinguishable: DINOv3’s Gram-anchored dense features and RADIO’s multi-teacher distillation each yield clean, semantically graded patch maps. The language-supervised encoders sit a step behind even with the text gate, CLIP at 0.902 and EVA-CLIP at 0.901. SAM (0.896) gives comparatively strong boundaries relative to its mIoU but weaker class discrimination. SigLIP consistently produced the lowest performance among the evaluated encoders, at 0.866, about three points below the next-weakest, its patch structure being the least suited to dense three-class prediction here. As a lightweight floor, the SegFormer head on DINOv3 already reaches 0.888 mIoU, indicating that DINOv3 features are already highly linearly separable for this task, and PyraFuse lifts that figure to 0.919, by recovering the boundary and thin-structure detail the flat head misses.

B. Loss Function Analysis

Table II ablates six objectives under the same roughly 16:1 class imbalance, all using the DINOv3 plus PyraFuse configuration [21]. Plain focal loss and class-weighted cross-entropy remain the weakest objectives for the skin class [22], with adjusted skin IoUs of 0.753 and 0.760, respectively. Focal loss appears to suppress the dominant easy-background contribution too aggressively, while weighted cross-entropy tends to over-segment skin near garment boundaries. Dice, plain cross-entropy, and Focal+Dice [21, 22] remain closely matched, achieving mIoUs of 0.919, 0.918, and 0.917, with corresponding skin IoUs of 0.825, 0.823, and 0.821. Pure Dice is marginally best overall, but the differences among the top three objectives are small and likely within normal run-to-run variation.

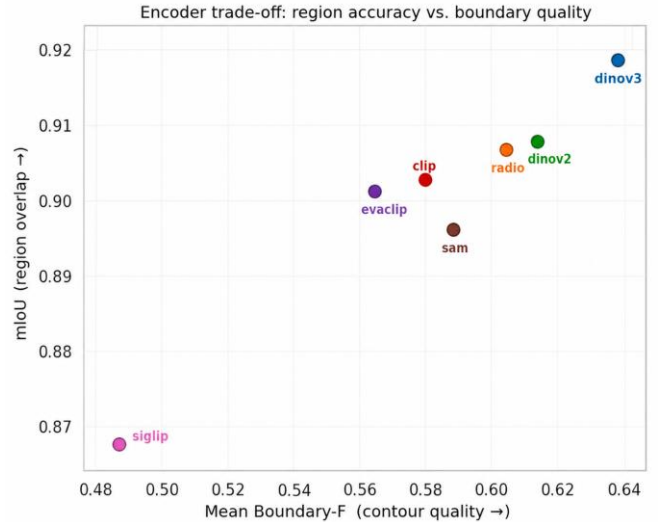


Fig. 6. Encoder trade-off from the 42-run benchmark: region accuracy (mIoU) against boundary quality (mean boundary-F). DINOv3 occupies the upper-right Pareto corner, with AM-RADIO and DINOv2 close behind and SigLIP trailing.

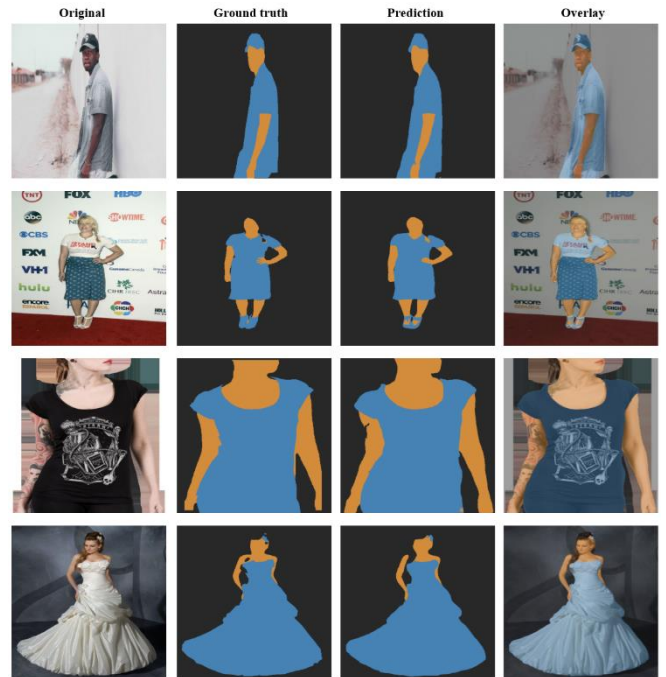


Fig. 7. Qualitative results on a random sample of test images.

TABLE I. Frozen-Encoder Comparison (Seven Backbones), Identical PyraFuse Decoder

Encoder	mIoU	mDice	IoU bg	IoU fab	IoU skin	Bdry-F
DINOv3	0.919	0.956	0.984	0.947	0.825	0.658
RADIO	0.907	0.950	0.981	0.936	0.805	0.604
DINOv2	0.906	0.949	0.981	0.937	0.802	0.610
CLIP	0.902	0.947	0.979	0.931	0.797	0.580
EVA-CLIP	0.901	0.946	0.979	0.929	0.795	0.565
SAM	0.896	0.943	0.977	0.922	0.790	0.590
SigLIP	0.866	0.925	0.968	0.891	0.747	0.494

TABLE II. Loss Function Ablation (DINOv3 + PyraFuse, Test Split)

Objective	mIoU	mDice	IoU bg	IoU fab	IoU skin
Dice	0.919	0.956	0.984	0.948	0.825
Cross-entropy	0.918	0.955	0.985	0.948	0.823
Focal+Dice	0.917	0.954	0.984	0.947	0.820
CE+Dice	0.911	0.951	0.982	0.944	0.808
Weighted CE	0.892	0.939	0.978	0.937	0.760
Focal	0.888	0.937	0.977	0.934	0.753

C. Backbone Size

The backbone-size ablation shows a consistent but diminishing improvement as encoder capacity increases. Under the identical PyraFuse and Dice training recipe, mIoU rises from 0.9136 with DINOv3-S to 0.9154 with S+, 0.9188 with B, and 0.9217 with L. The gains are particularly visible for the more difficult skin class and at object boundaries, where skin IoU improves from 0.8160 to 0.8292 and mean boundary F-score from 0.6403 to 0.6705. However, these improvements come with a substantial increase in model size, from 22.7M total parameters for S to 304.5M for L. DINOv3-B therefore provides a favorable accuracy-capacity trade-off, recovering most of the performance of the much larger L backbone while requiring roughly one-third of its parameters. The pretrained models used in this comparison are publicly available <https://huggingface.co/jamal-one/PyraFuse>.

D. Regularization and Schedule

Because the trainable head is small relative to some 44,783 images, the main risk is feature-space shortcut learning (e.g., tying skin to particular hues) rather than classical overfitting. MixUp (alpha = 0.2) [23] and CutMix (alpha = 1.0) [24], each applied with probability 0.5 per batch, counter this tendency; stochastic depth and spatial dropout regularize the decoder; and weight EMA stabilizes the endpoint. Training uses AdamW [25] with learning rate $3e-4$ and weight decay $1e-4$, cosine decay after a 2-epoch linear warmup, gradient clipping at 1.0, a large batch at 448 by 448, and 100 epochs, with validation mIoU on the EMA weights serving as the selection criterion. A complete run finishes in about 6.3 hours on a single GPU, roughly 375 seconds per epoch, and reaches its best validation mIoU of 0.929 at epoch 52.

VI. FAIRNESS AND ROBUSTNESS

A. Skin-Tone Fairness Audit

Stratifying the DINOv3 + PyraFuse audit over the 4,341 test images that contain skin (from 4,594 total) reveals a real and reportable gap. Skin IoU is 0.732 for the darkest-tone group, 0.804 for the medium group, and 0.815 for the lightest group, with a per-image mean of 0.784 (Fig. 8). The worst group, the darkest, trails the mean by about five points and the lightest group by about eight. When the same audit is binned into six Fitzpatrick-proxy bands, the darkest band, Type VI, drops to 0.688 IoU, roughly 11 to 14 points below the mid-tone bands.

This trend is consistent with observations reported in prior studies of skin and face analysis systems, and may reflect the

under-representation of darker skin in the visuAAL annotation distribution. Tone-preserving augmentation narrows this gap but does not close it. Two limits of scope should be noted: the audit covers the single deployed model rather than all seven encoders, and the lightness proxy, median HSV value, approximates rather than directly measures the Monk Skin Tone scale [26]. We include these results because worst-group skin recall is particularly relevant for privacy-oriented deployments; we therefore report the per-group gap directly rather than only the mean.

B. Qualitative Results on Hard Cases

Figure 9 shows predictions on demanding test cases that include skin-colored garments, sheer fabric, occluding hands, and low light. PyraFuse holds accurate boundaries through these situations, distinguishing skin from similarly colored fabric and coping with partial occlusion. The lightweight decoder retains fine detail around fingers, neckline, and fabric edges that heavier heads sometimes smooth away. Notably, in several cases where the ground-truth annotation is itself incomplete, the model generalizes beyond the provided labels and recovers the missing regions, producing more complete and coherent segmentations than the reference masks.

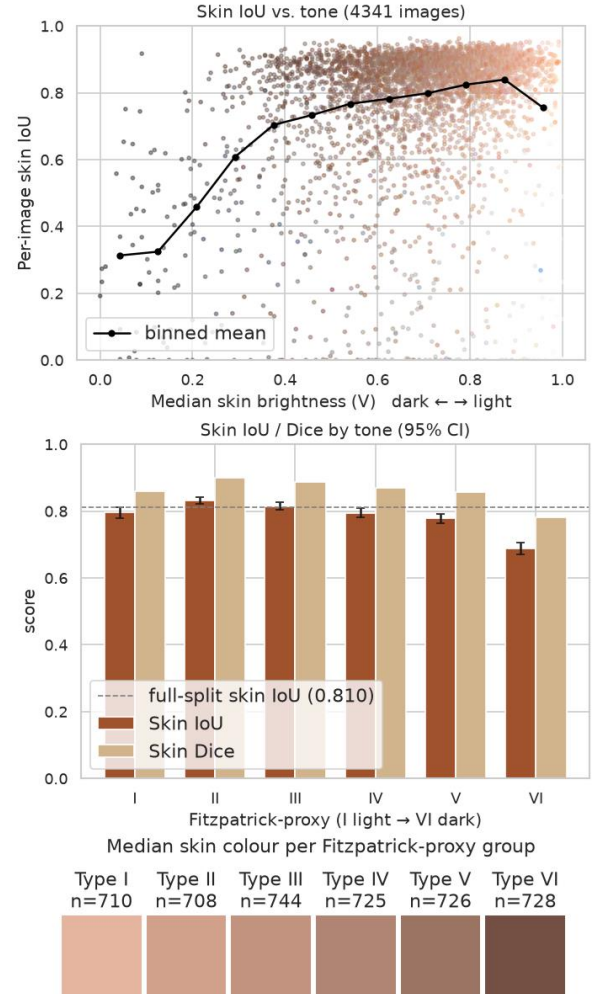


Fig. 8. Skin segmentation performance stratified by Fitzpatrick-proxy tone groups.

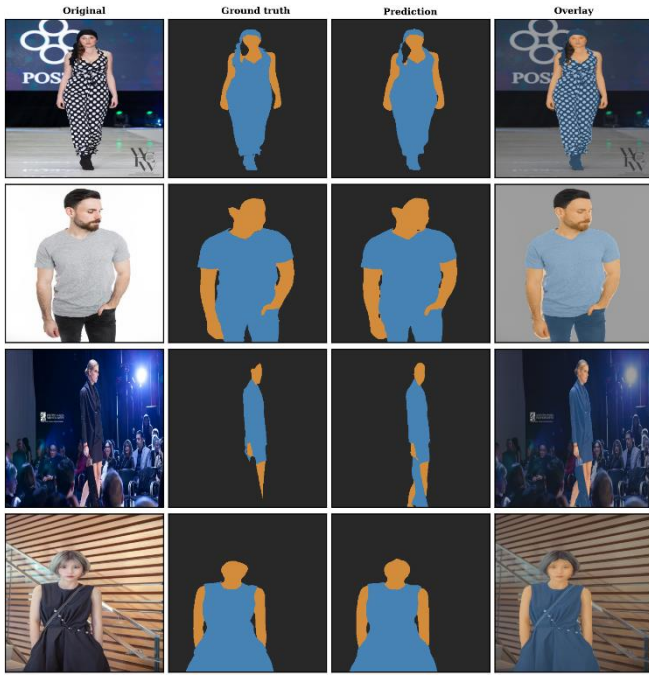


Fig. 9. Qualitative results on hard cases: skin-colored garments, sheer fabric, occluding hands, and low light. Columns show input, ground truth, prediction, and overlay.

VII. REAL-TIME DEPLOYMENT

The system achieves real-time performance through a two-stage TensorRT pipeline in which the DINOv3 encoder and decoder are exported separately to ONNX and compiled independently. This allows flexible placement and precision selection, as well as reuse of pre-computed encoder features. Logits are upsampled on the GPU before argmax, with optional guided filtering for edge refinement [27] and adaptive temporal EMA to suppress frame-to-frame flicker.

As shown in Fig. 10, TensorRT substantially improves throughput across all backbone sizes. FP16 preserves essentially the same segmentation accuracy as eager FP32 while providing a large speedup. For the smaller backbones, throughput increases from roughly 130 to 140 img/s in eager mode to about 190 to 200 img/s with TensorRT FP32, and to around 320 img/s with mixed precision. INT8 provides the highest throughput, reaching approximately 340 to 350 img/s, but consistently reduces mIoU by about 0.5 percentage points. The same trend holds for larger backbones, with the L model reaching about 140 to 150 img/s in mixed/INT8 TensorRT compared with only about 30 img/s in eager FP32. Overall, mixed-precision TensorRT provides the best accuracy-throughput trade-off, while INT8 represents the maximum-throughput configuration at a modest accuracy cost.

VIII. CONCLUSION

We have presented a practical, deployment-oriented framework for joint skin-fabric semantic segmentation. The merged-dataset procedure brings visuAAL skin masks and Fashionpedia garment instances together into one three-class corpus, and the seven-VFM frozen-encoder benchmark indicates DINOv3, with AM-RADIO and DINOv2 close behind, is the strongest backbone and SigLIP as the weakest. PyraFuse offers a favorable accuracy-to-computation trade-

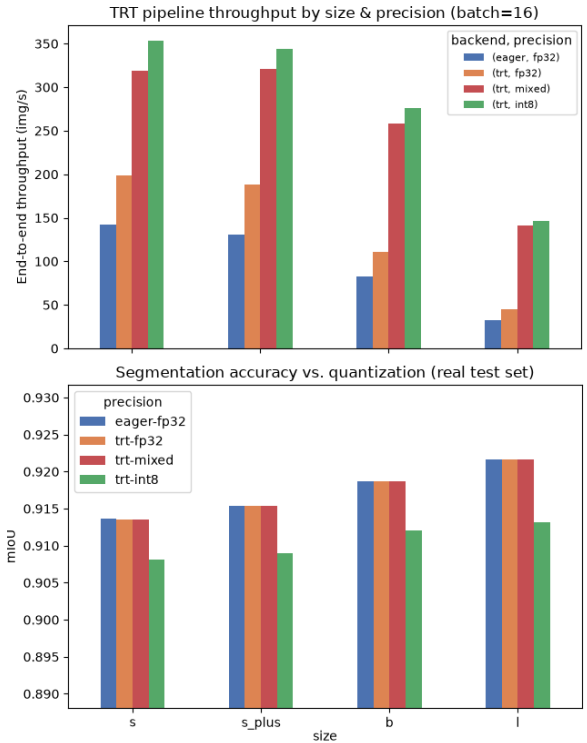


Fig. 10. TensorRT precision and throughput across different backbone sizes

off in our benchmark. The loss study finds Dice, cross-entropy, and Focal+Dice essentially equal, and we retain Dice for its soft-target compatibility. A direct skin-tone fairness audit exposes a residual deficit of roughly five points on the darkest group relative to the mean, a limitation we report as open rather than solved. Finally, an end-to-end TensorRT showcase demonstrates 300 images per second at FP16 with negligible accuracy loss, and the frozen-encoder plus lightweight-decoder design exports cleanly, which validates the approach as a deployment decision as much as a training one. Future work includes distilling the frozen encoder into a mobile-class student, extending the label space to accessories and hair, and exploiting the large pool of unlabeled fashion images through semi-supervised learning.

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